

Technical Mentor Report

McCaskill Algorithm Optimization Using AlphaZ

Fall 2015

Research Mentor: Swetha Varadarajan

Research Group: Prof. Rajopadhye’s MELANGE Research Group, Colorado State University¹

Institution: Colorado State University

Overview

I joined Professor Sanjay Rajopadhye’s MELANGE Research Group at Colorado State University in January 2014 as a graduate research assistant. My research focused on compiler optimization of dynamic programming algorithms arising in computational biology, with particular emphasis on RNA–RNA interaction computations using polyhedral compilation techniques [2].

During the Fall 2015 semester, I served as the technical mentor for a graduate research project investigating compiler optimization of the McCaskill RNA partition function algorithm [1] using the AlphaZ polyhedral compilation framework [3].

Mentoring Activities

As the project mentor, my responsibilities included:

- Introducing the McCaskill RNA partition function algorithm and its dynamic programming formulation.
- Explaining the computational complexity and dependence structure of the algorithm.
- Guiding dependence analysis of computational kernels.
- Assisting with affine kernel extraction and AlphaZ implementation.
- Discussing polyhedral scheduling, compiler optimization, and parallelization strategies.
- Reviewing implementation progress and performance evaluation.

The project provided practical experience in applying compiler optimization techniques to recursive dynamic programming algorithms used in computational biology.

¹Professor Sanjay Rajopadhye’s homepage: <https://www.cs.colostate.edu/~svr/>.

Research Context

The mentoring activities were carried out alongside my graduate research on compiler optimization of RNA–RNA interaction algorithms [2]. Both efforts shared common research themes, including dependence analysis, affine transformations, polyhedral scheduling, and parallel execution of recursive dynamic programming algorithms.

The project contributed to the broader research activities within the MELANGE Research Group on compiler optimization for computational biology applications.

References

- [1] J. S. McCaskill, “The Equilibrium Partition Function and Base Pair Binding Probabilities for RNA Secondary Structure,” *Biopolymers*, vol. 29, no. 6–7, pp. 1105–1119, 1990. doi: <https://doi.org/10.1002/bip.360290621>.
- [2] S. Varadarajan, *Polyhedral Optimizations of RNA–RNA Interaction Computations*, M.S. Thesis, Department of Computer Science, Colorado State University, Fort Collins, Colorado, 2017. ProQuest Dissertations & Theses, Publication No. 10690938.
- [3] T. Yuki, G. Gupta, D. Kim, T. Pathan, and S. Rajopadhye, “AlphaZ: A System for Design Space Exploration in the Polyhedral Model,” In *Proceedings of the 25th International Workshop on Languages and Compilers for Parallel Computing (LCPC)*, Lecture Notes in Computer Science, vol. 7760, Springer, 2013, pp. 17–31.